Lab 02 Manual

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SRT411NAA: Digital Data Analytics

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Date Submitted: May 28th, 2024.

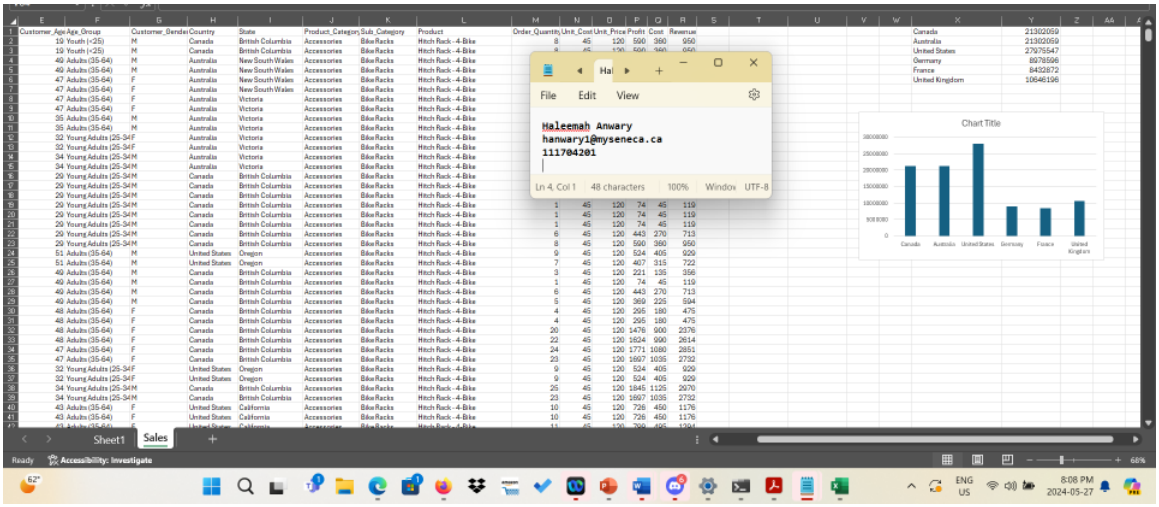
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Select Appropriate Data Features for Chart Elements (Task 01)

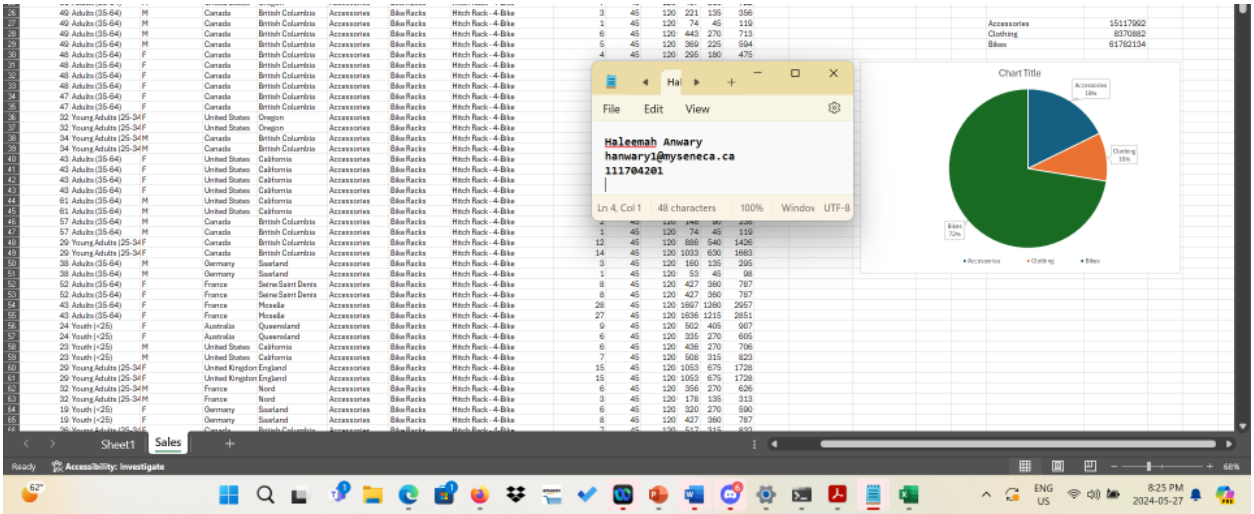
Bar Chart: Revenue by Country

Explanation: Here, I am using formulas to create the columns I need, the formula =UNIQUE(H2:H113037) will give us a unique list of countries and the next formula =SUMIF(H:H, X1, R:R) will give us the total revenue for the country. This bar chart shows revenue by country. Bar charts are ideal for comparing several items across one categorical variable. Country-based revenue can visually represent which countries generate more or less revenue, providing a clear comparison.



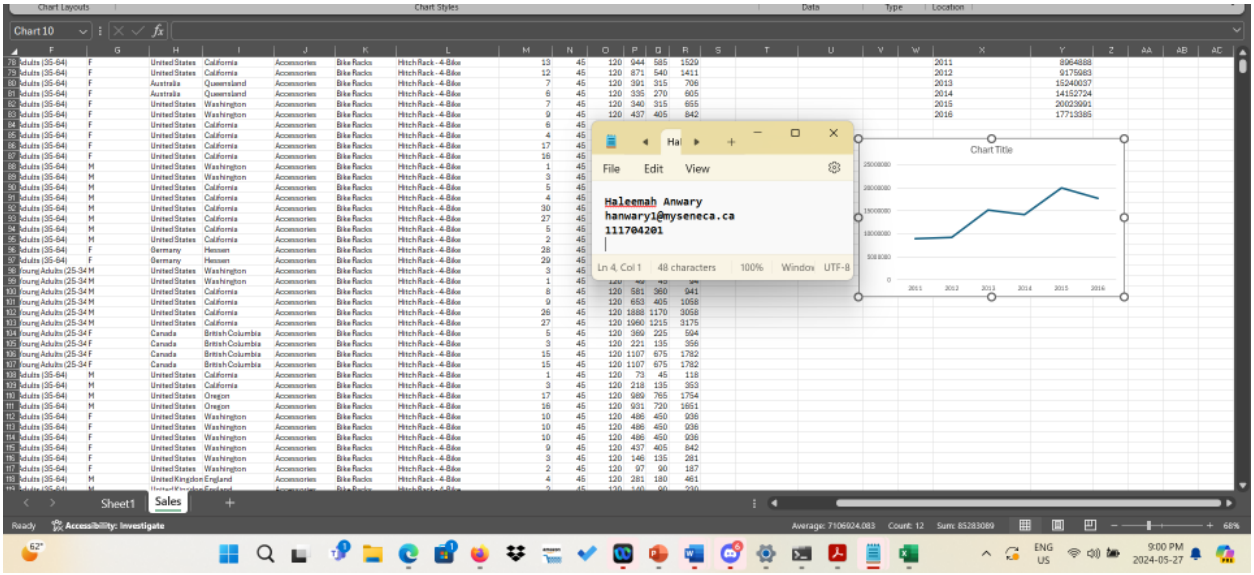
Pie Chart: Sales Percentage by Product Category

Explanation: Here, like the Bar chart example I'm using the formulas to extract the different columns and calculate the revenue. This pie chart shows Sales Percentage by Product Category. Pie charts are suitable for displaying the proportional distribution of categories. Using the Product Category for a pie chart allows us to see the market share of each category.



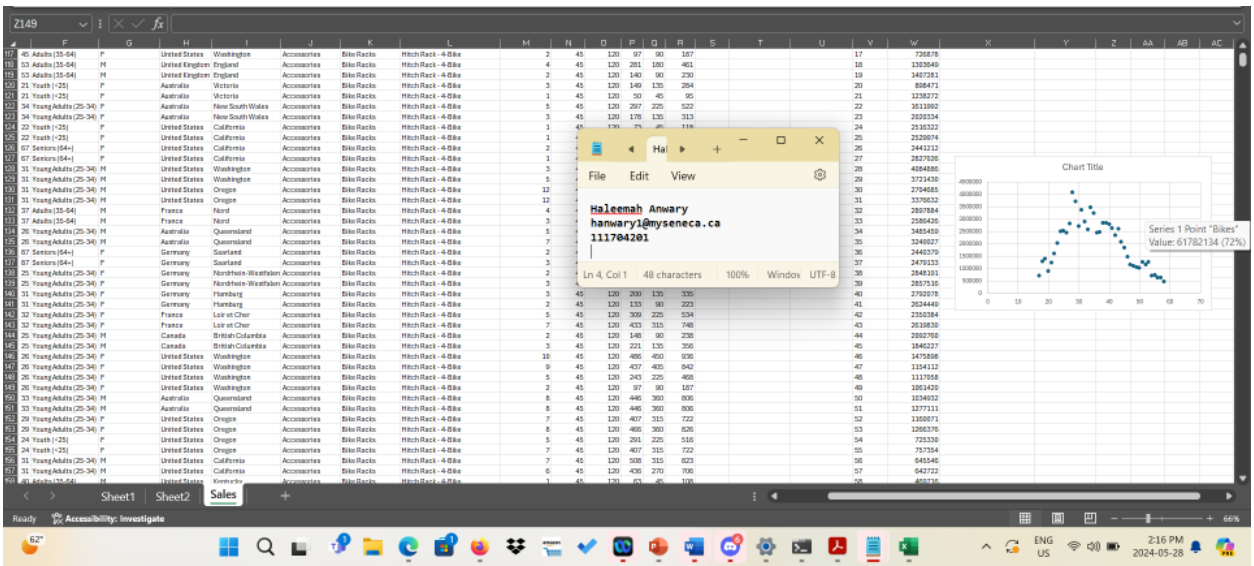
Line Chart: Yearly Revenue

Explanation: Here I used the formula =SORT(UNIQUE(D2:D113037)) to sort the years in order and used the =SUMIF(D:D, X78, R:R) to get the total yearly revenue. Line charts are great for displaying data changes over time, allowing us to identify trends, peaks, and troughs in revenue across different time frames.



Scatter Plot: Customer age and revenue generated.

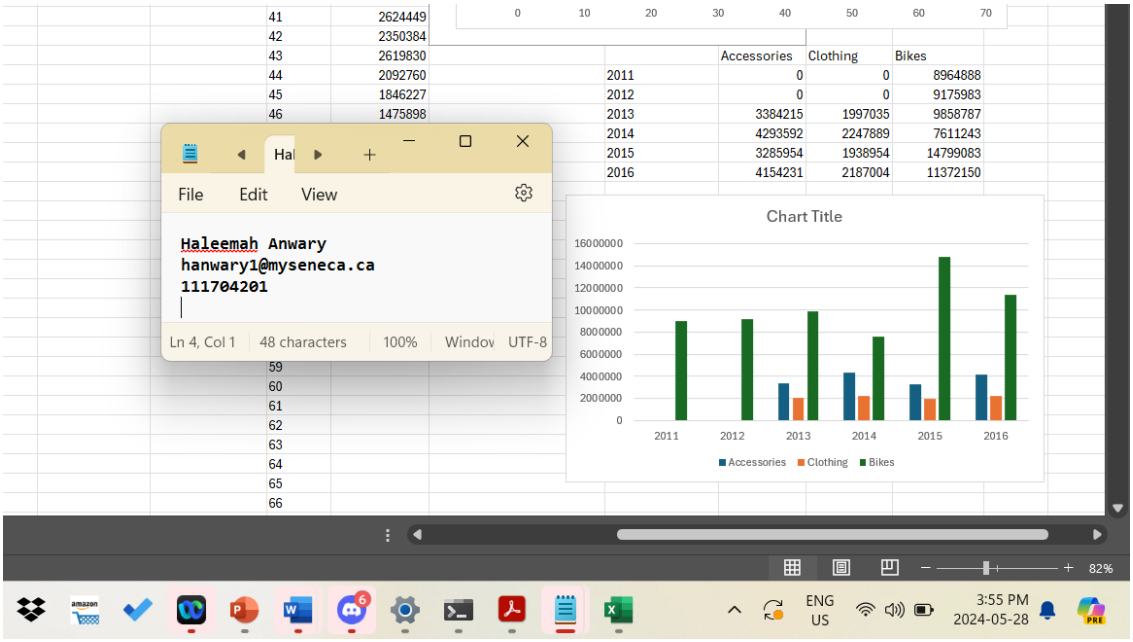
Explanation: Scatter plots are used to observe relationships between two variables. Here, plotting 'Customer_Age' and 'Revenue' can help identify if there's any correlation between the age of customers and the revenue they generate.



Stacked Bar Chart:

Compare the revenue from different product categories over multiple years.

Explanation: Stacked bar charts are useful for comparing multiple data series in one diagram. It allows for seeing the total revenue per year, as well as the contribution of each product category to that total.

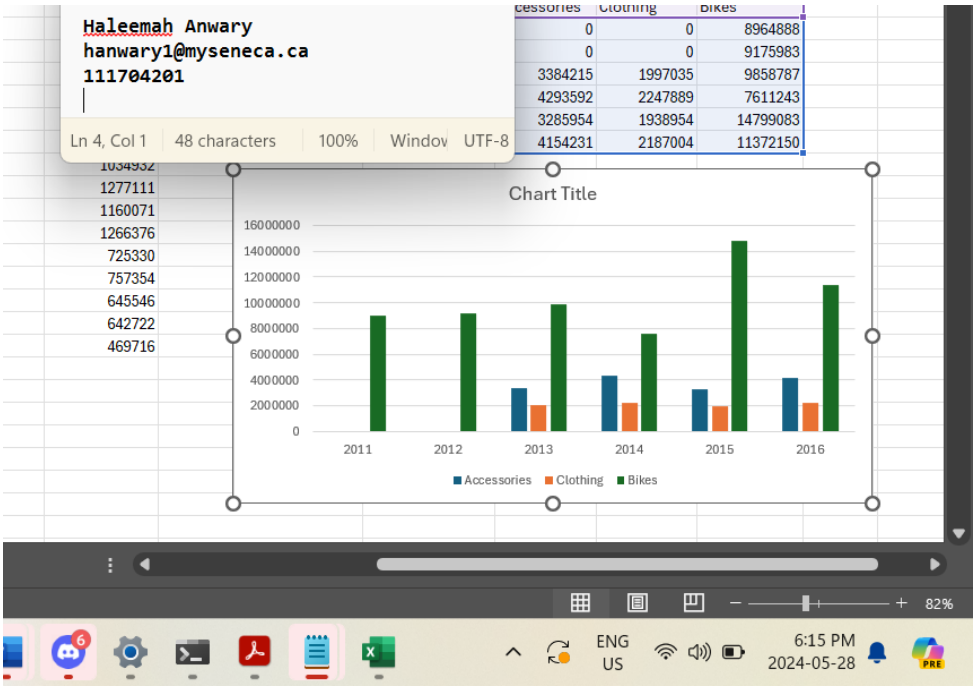


Using the Switch Row/Column Feature (Task 02)

Switching rows and columns alters the focus of the analysis significantly. It changes the narrative from a time-based progression story to a categorial analysis across time. The new representation retains its relevance but serves a different analytical purpose.

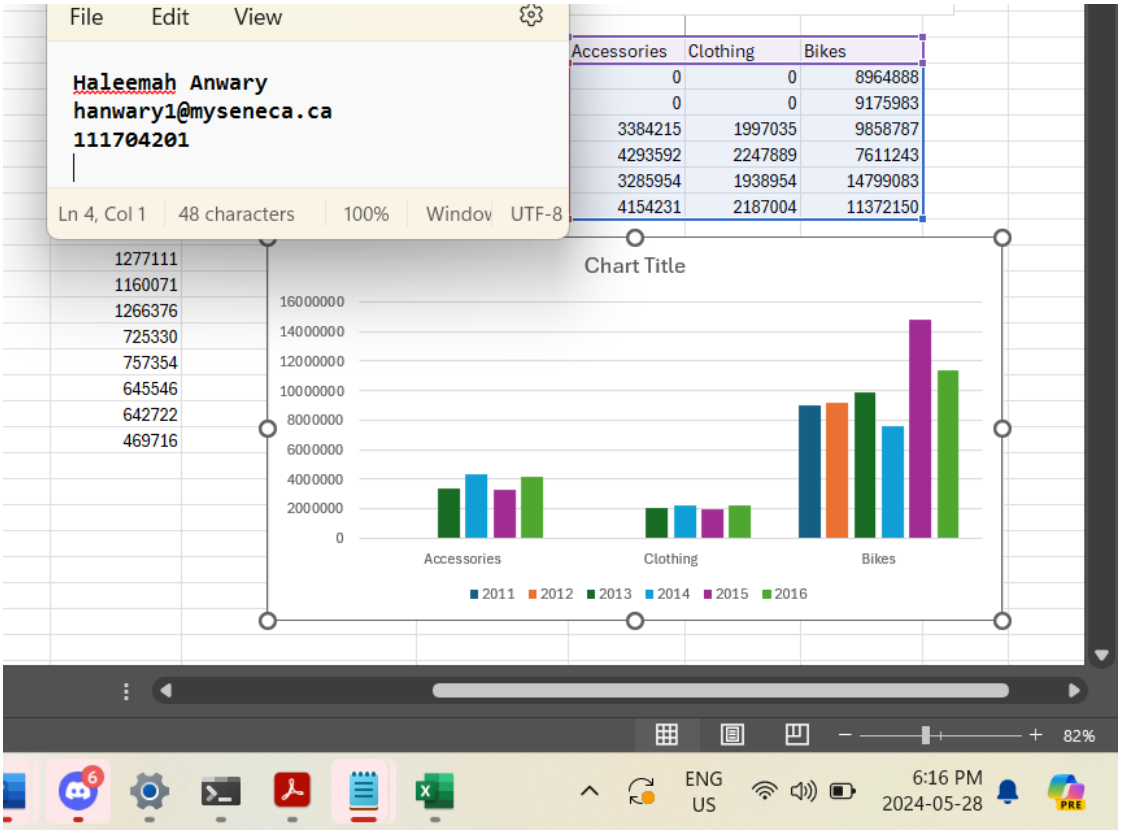
Using the Switch Row/Column Feature on the Stacked Bar chart.

As we can see in the original stacked bar graph, the bars represent a year with segments stacked to show revenue from different product categories within that year. The horizontal axis has the years, and the vertical axis shows the revenue.



After clicking on the Switch Row/Column

This action will change how the data is organized in the chart. The horizontal axis will now have the product categories and each bare now represents a product category. The segments in each bar now represent different years, showing the revenue contribution from each year within a single product category.



Is it still effective?

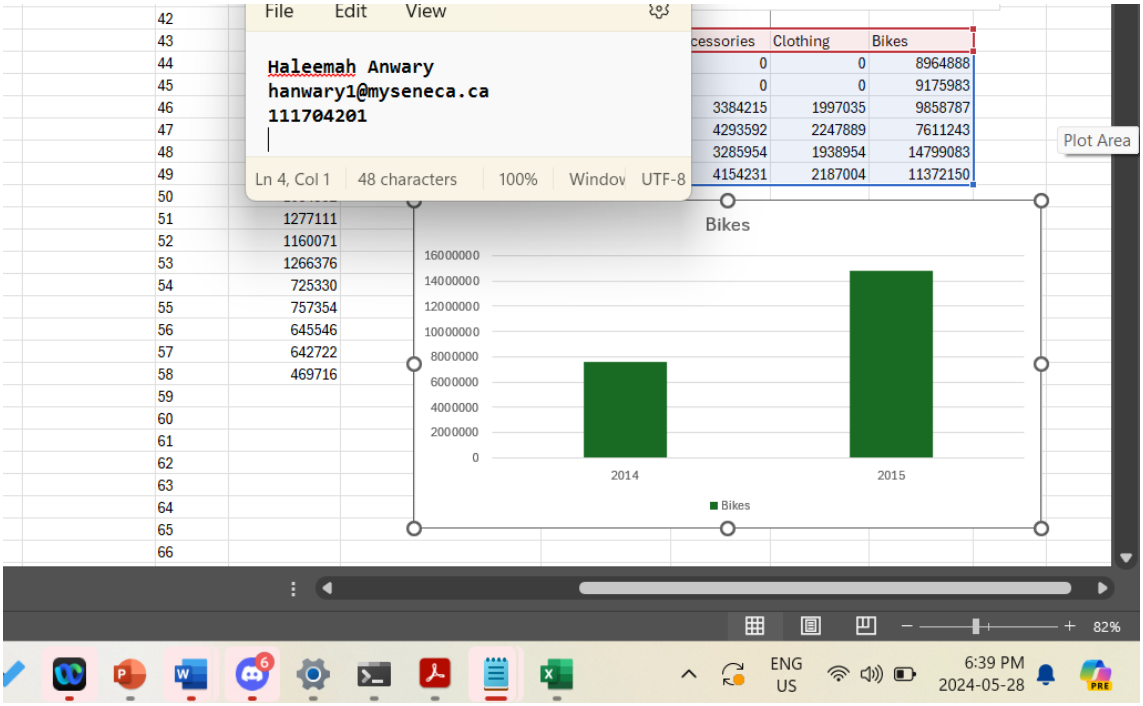
The new format can be useful if the primary interest is to analyze and compare the performance of each product category across multiple years. If the intent was to track changes in total revenue year over year and see how different categories contributed to these changes, the new layout might be less effective.

Utilising Chart Filters (Task 03)

We can utilize the 'Chart Filters' button next to the upper right corner of the chart. It often appears as a funnel icon. Once we click on it, we can see options to filter by 'Legend Entries (Series)' and 'Horizontal (Category) Axis Labels.'

Applying Filter to the Stacked Bar Chart

To focus on a particular year, I deselect all other years I don't need in the 'Horizontal (Category) Axis Labels.' This action will leave the bars representing only the selected year visible on the chart. Alternatively, to highlight a specific product category, you could deselect all other categories under the 'Legend Entries (Series)' section.



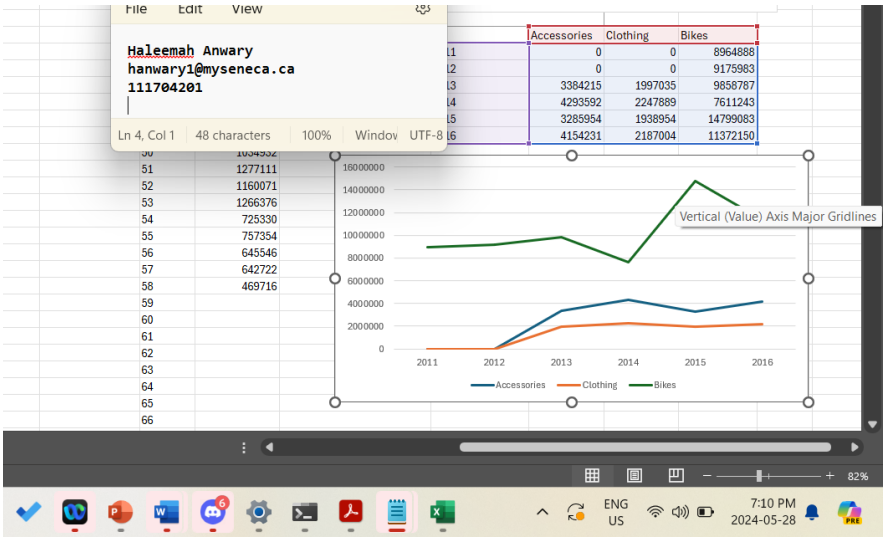
Why Utilize Chart Filters

Initially, the chart displayed a comprehensive view with all years and product categories shown. This setup is ideal for a broad analysis but can be overwhelming when specific insights are needed. With filters applied, the chart now only shows the data for selected years or categories. This focused view eliminates clutter and allows analysts and stakeholders to concentrate on the specifics of interest.

Changing Chart Types (Task 04)

Stacked Bar Chart Changed to Line Chart

Each product category now has its own line on the chart, with the x-axis still representing years and the y-axis representing revenue. Each point on a line shows the revenue from a product category in a particular year. Line charts are excellent for showing trends over time, this makes it easier to observe the growth or decline in revenue for each product category across the years.



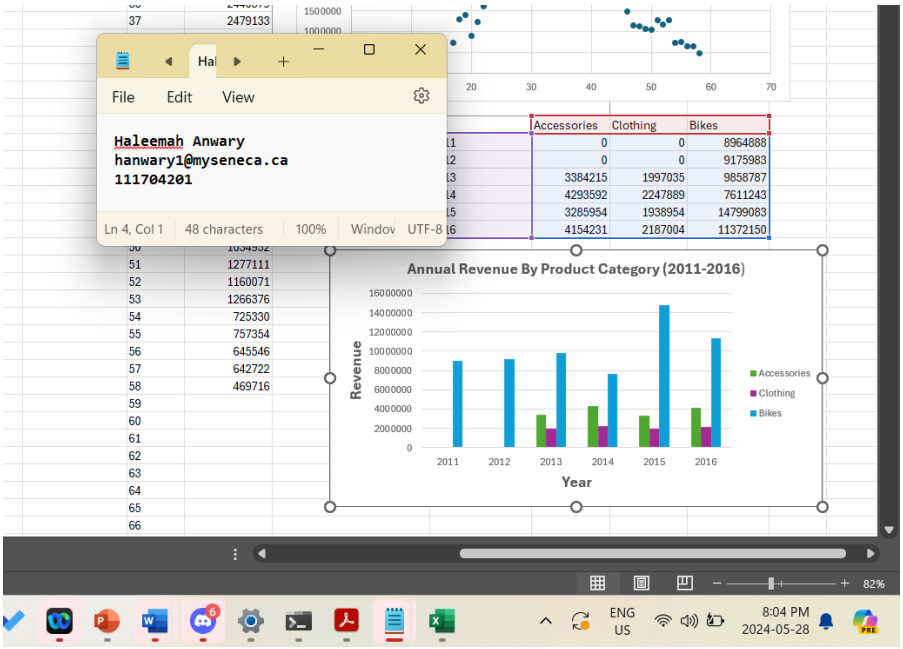
Is it still effective?

However, there is loss of composition and overlap issues if the product categories had similar revenue paths. If the goal is to analyze the trend of revenue over time for each category independently, the line chart is great. On the other hand, if the aim was to analyze the contribution of each category to the overall revenue within each year, this insight is less clear without the cumulative visual affects of the stacked bar chart.

Format and Enhance Chart Visual Appeal (Task 05)

Enhanced Stacked Bar Chart

- **Changed the bar colors:** To differentiate each product category clearly.
- **Added title and labelled X-axis and Y-axis:** To provide clear, informative data points without clutter.
- **Changed font and made bold:** Improving readability and ensuring the chart is easily legible even from a distance.

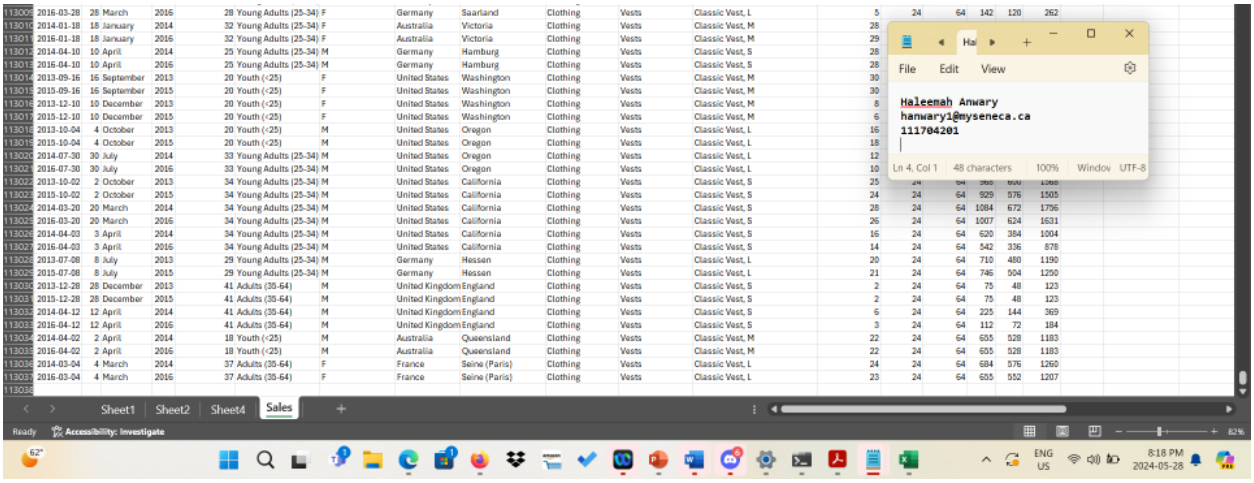


Importance of Well-Presented Charts in Data Storytelling

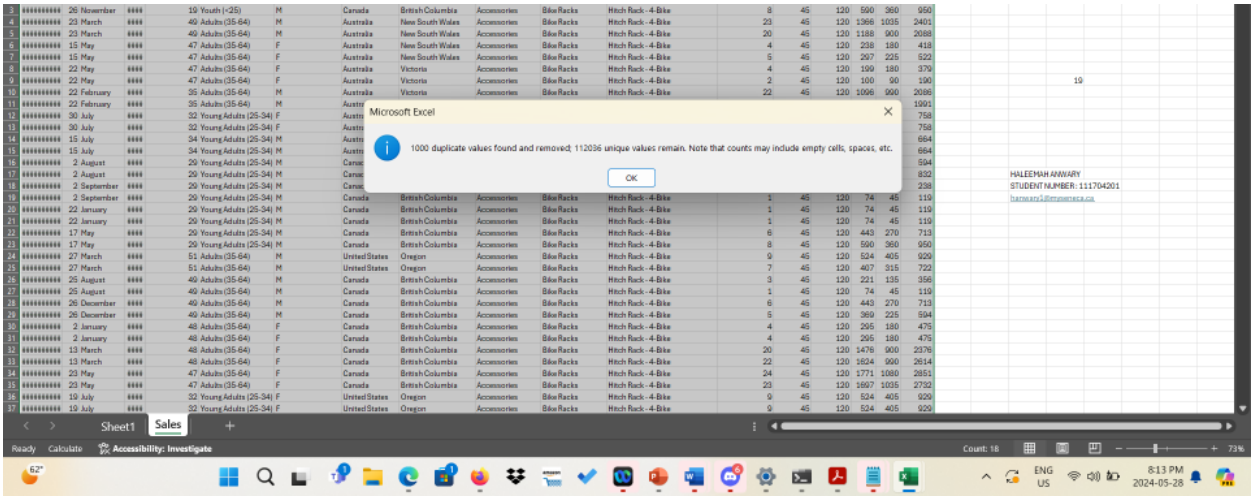
Well-presented charts in data storytelling are important as it enhances engagement, understanding, professionalism, and decision making. By carefully formatting the stacked bar chart, we ensure that it not only carries the right information but also presents it in the most effective way possible. This enhances the chart's role in data storytelling by making it both visual and informational asset.

Remove Duplicates (Task 06)

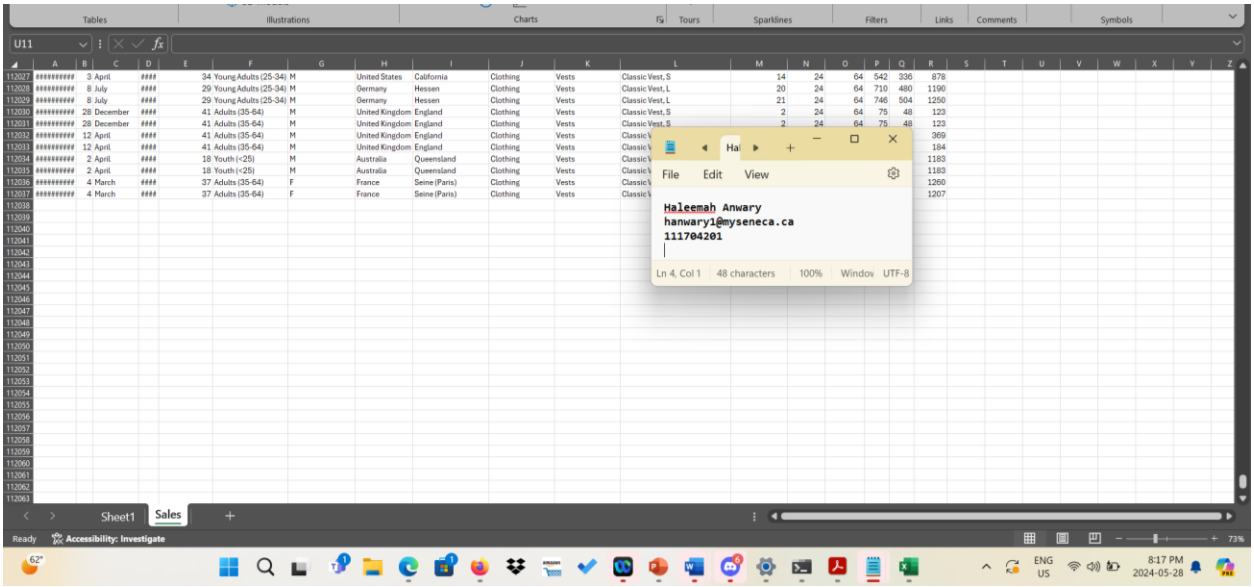
Before Duplicate Removal: There are 113036 values.



Removed: After going to data and removing duplicates, the message from Microsoft Excel shows that 1000 duplicate values have been removed.



After Duplicate: There are 112036 values remaining.

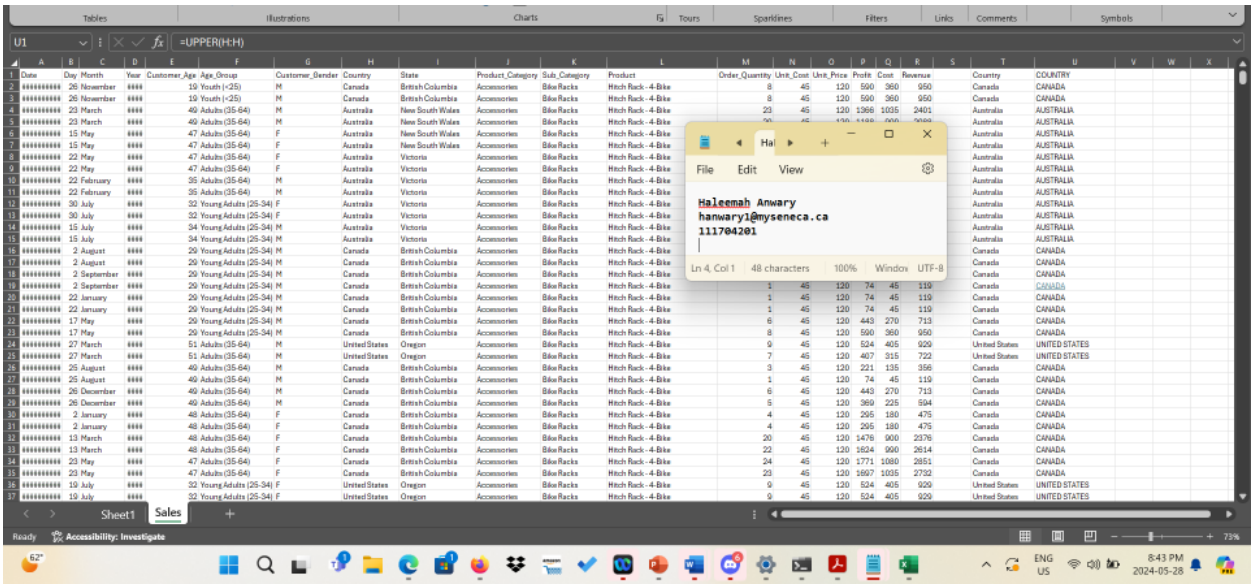


Importance of Removing duplicates

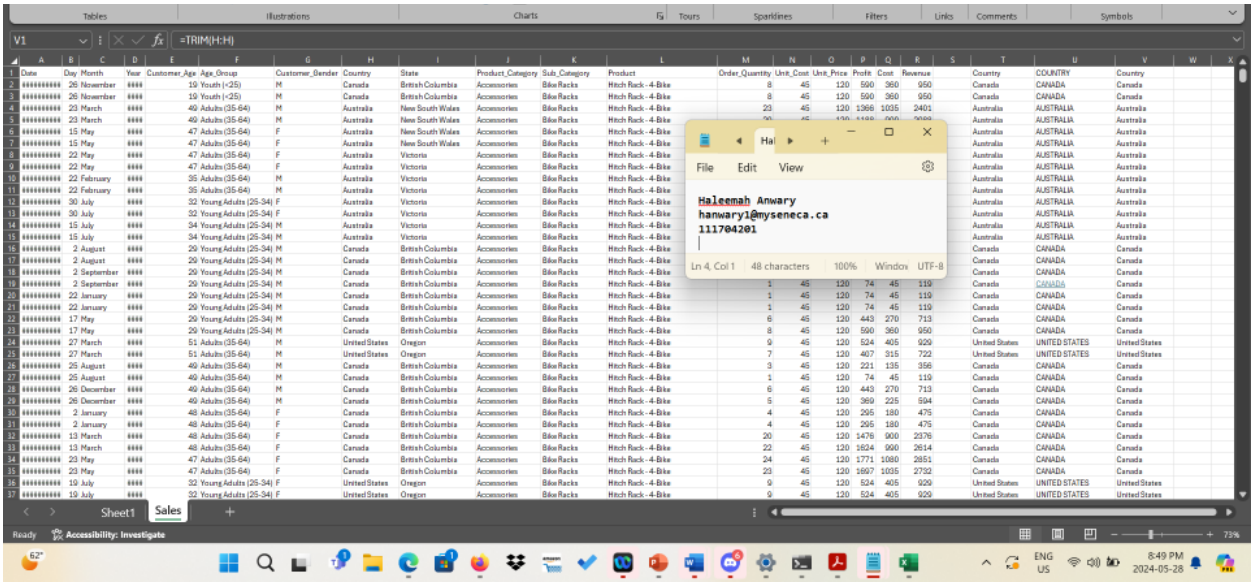
It's important to remove duplicates to help in achieving more reliable and valid insights, fewer data points can result in faster processing times for data analyses, and it ensures that metrics such as sums, average, and counts are calculated based on unique entries, providing a true representation of the data.

Standardise Data Using Formulas (Task 07)

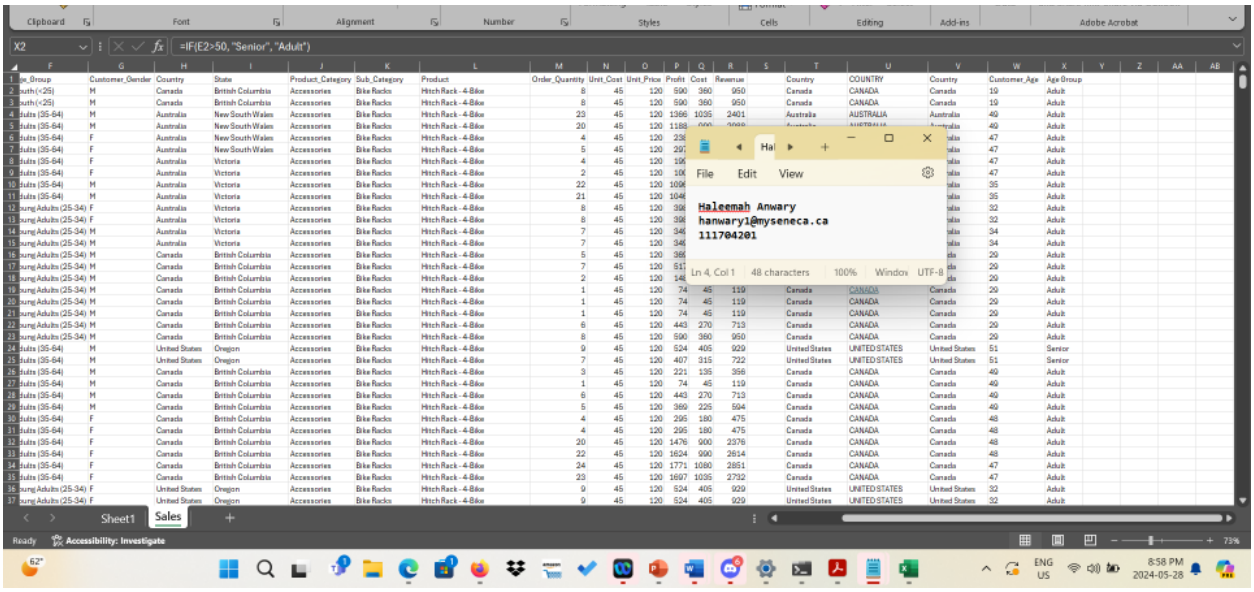
Standardized Country names with the formula =UPPER(H:H)



Remove extra spaces from the country column, this is demonstrated in Column V with the formula =TRIM(H:H).



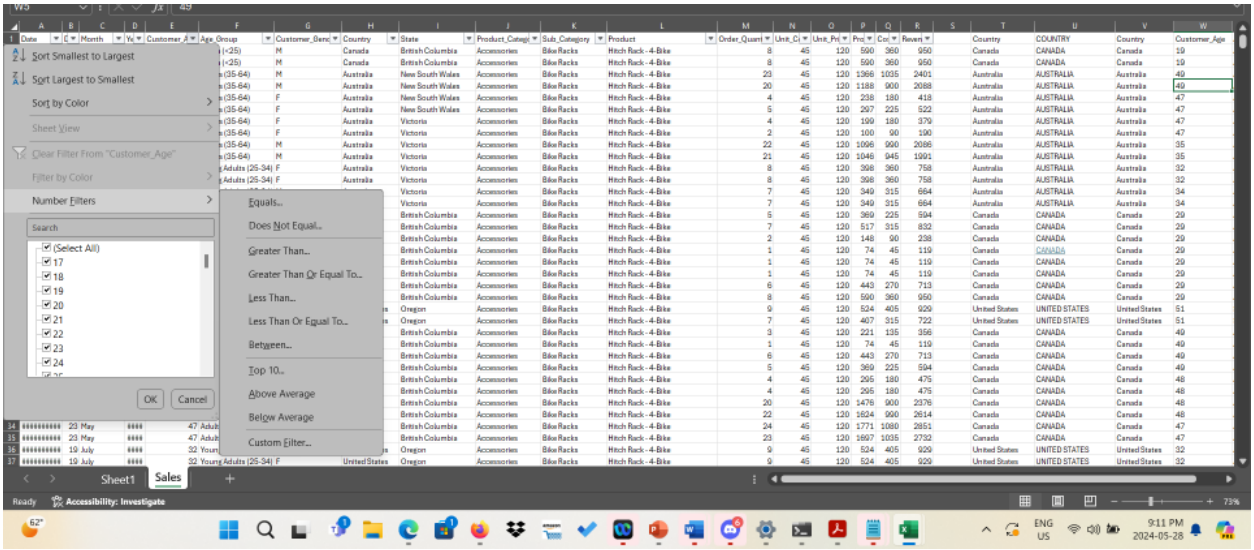
Categorize Age groups: I used the formula =IF(E2>50, "Senior", "Adult")



Using Data Filters (Task 08)

You can filter based on specific criteria. For instance:

- **Text Filters:** For columns like "Country", use text filters to isolate unusual or misspelled entries.
- **Number Filters:** For numeric columns like "Age" or "Revenue", filter for exceptionally high or low values that might indicate data entry errors.
- **Custom Filters:** Use custom filters to find entries that don't match expected patterns, such as non-standard age groups in the "Age_Group" column.



Importance of Filtering

Excel's Data Filter feature boosts efficiency by allowing quick isolation of specific data segments, eliminating the need to manually sift through extensive datasets. This targeted approach notifies you of any errors, improving the accuracy of your data which is essential for trustworthy analysis. Additionally, filtering helps in promptly identifying anomalies or outliers that might skew results, ensuring the integrity of your data analysis.

Feature Engineering (Task 09)

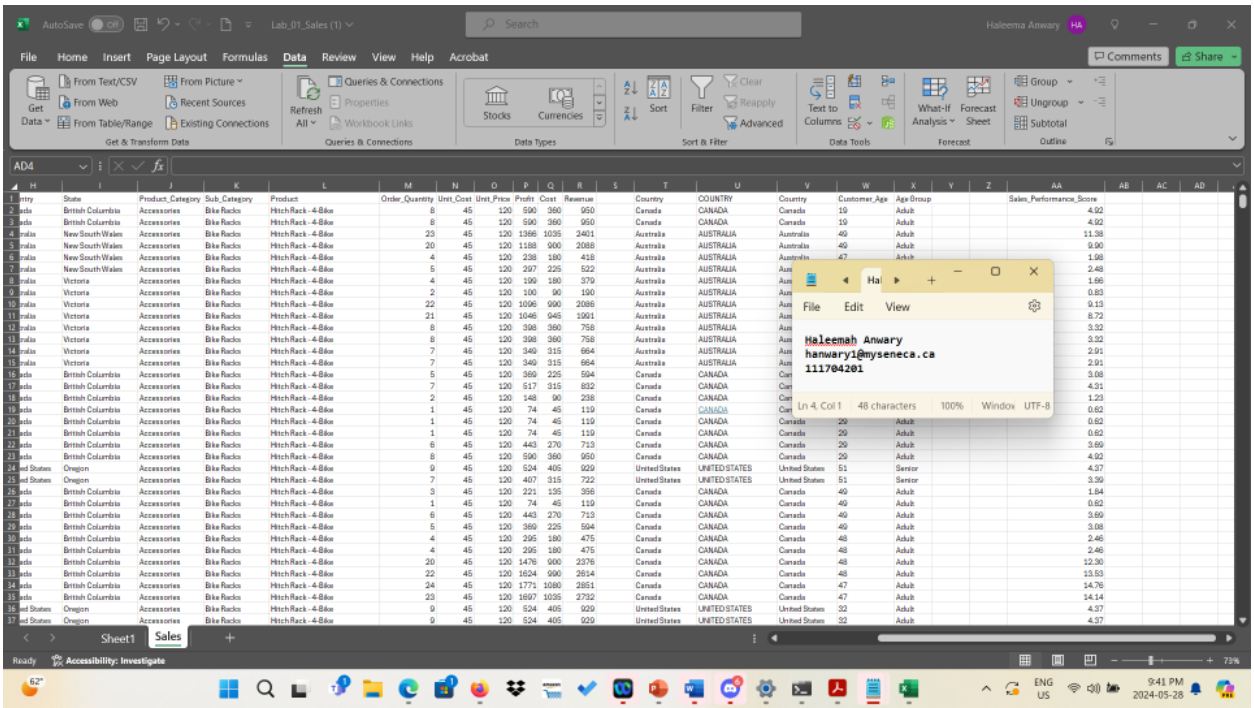
Creating a Sales Performance Score Feature

Data Needed: Utilize the 'Profit', 'Unit_Price', and 'Unit_Cost' columns.

I used a formula to calculate a score that factors in the profit margin and the ratio of the selling price to the cost price: =IFERROR((Profit / Unit_Cost) / (Unit_Price / Unit_Cost), 0).

Use Case: This new metric can help identify which products are yielding high returns relative to their cost and pricing strategy.

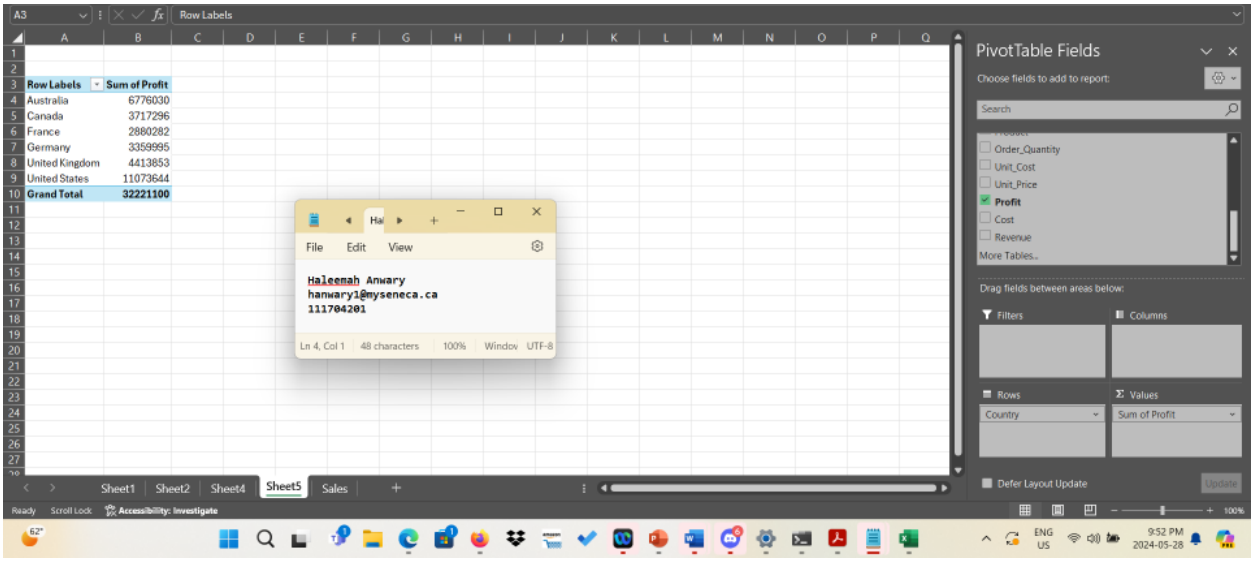
Benefit: Without this feature, analysis might focus solely on gross profit or revenue, which doesn't account for the relative cost or pricing efficiency. With the 'Sales_Performance_Score', stakeholders can now evaluate the efficiency of sales in generating profit relative to their pricing and cost structure, providing deeper insights into the operational effectiveness.



Create Pivot Tables (Task 10)

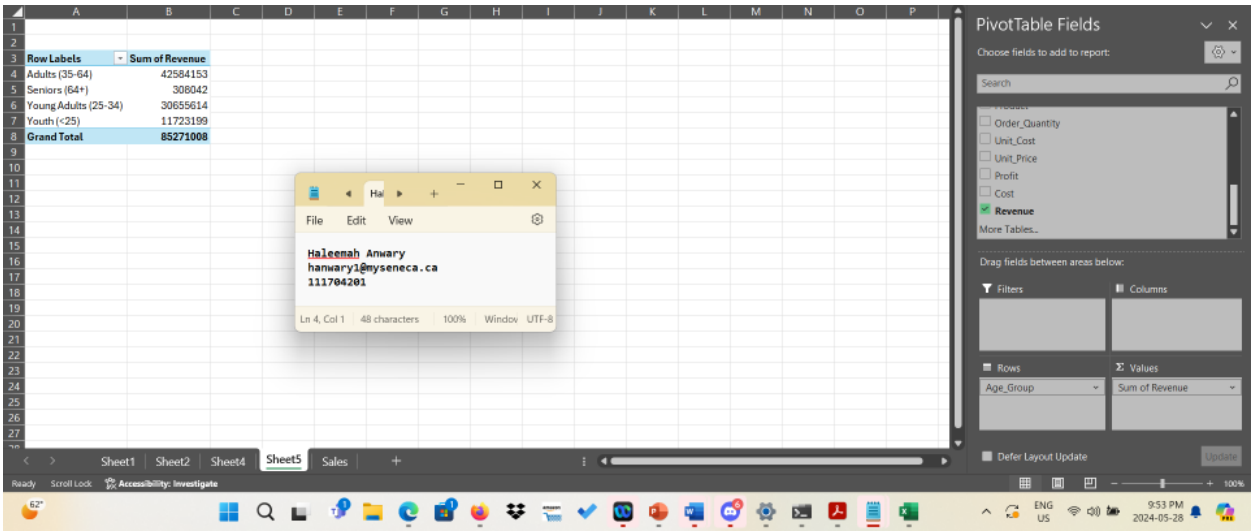
Story 1: Highest Profit by Country

Identify which country has the highest total profit.



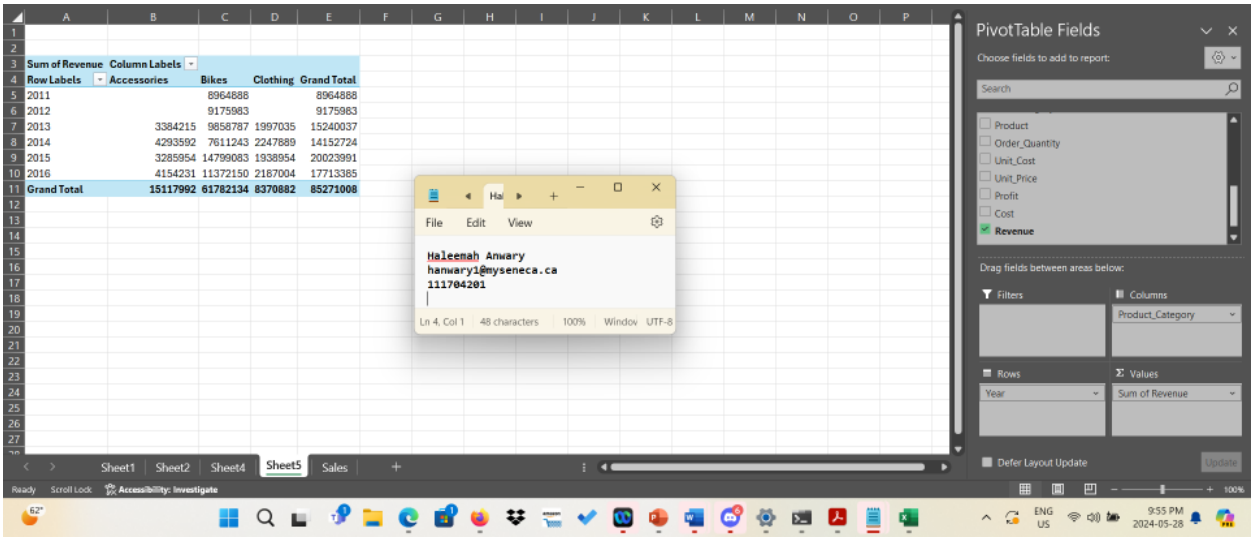
Story 2: Highest Sales by Age Group

Determine which age group generates the highest revenue.



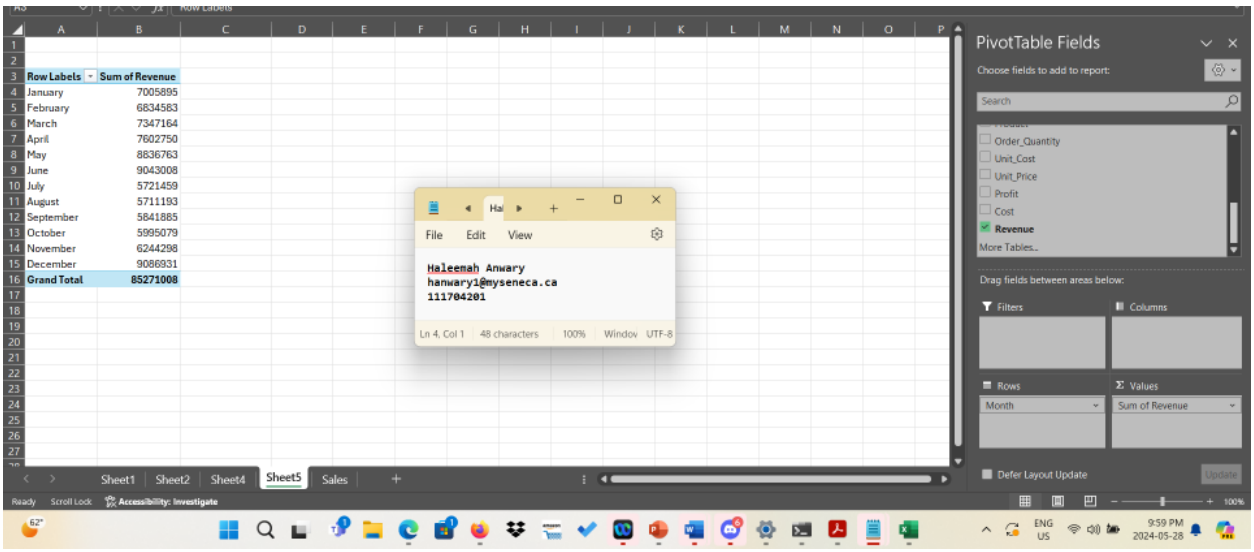
Story 3: Product Category Performance by Year

Analyze how different product categories perform each year.



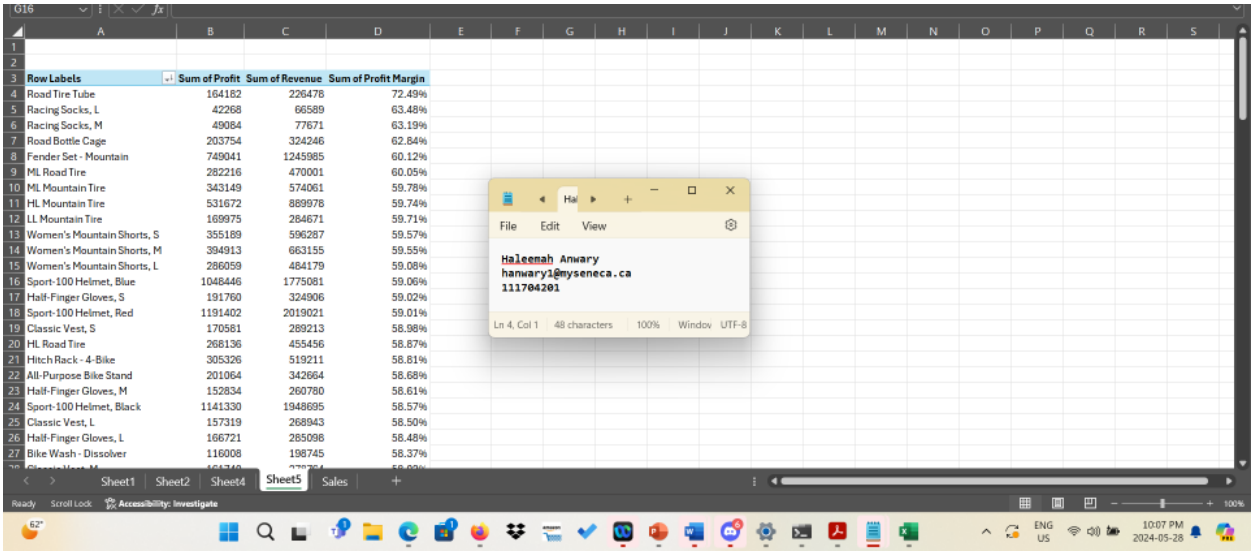
Story 4: Monthly Revenue Trends

Understand how sales revenue trends vary month by month.



Story 5: Profit Margin Analysis by Product

Identify which products have the highest profit margins.

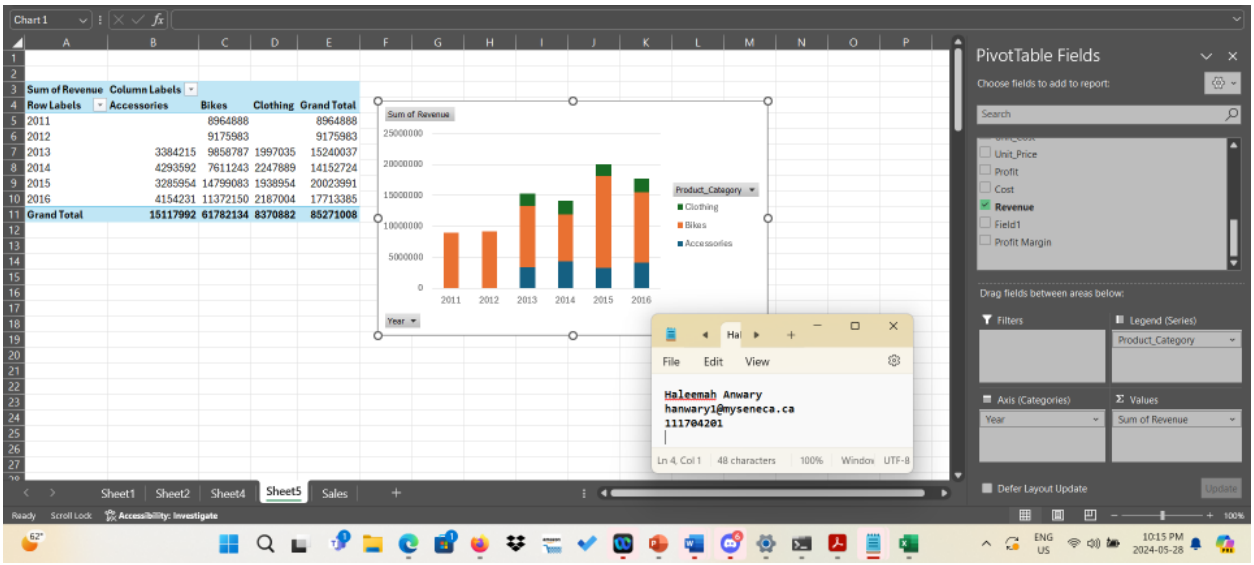


Row Labels	Sum of Profit	Sum of Revenue	Sum of Profit Margin
Road Tire Tube	164182	226478	72.49%
Racing Socks, L	42268	66589	63.48%
Racing Socks, M	49084	77671	63.19%
Road Bottle Cage	203754	324246	62.84%
Fender Set - Mountain	749041	1245985	60.12%
ML Road Tire	282216	470001	60.05%
ML Mountain Tire	343149	574061	59.78%
HL Mountain Tire	531672	889978	59.74%
LL Mountain Tire	169975	284671	59.71%
Women's Mountain Shorts, S	355189	596287	59.57%
Women's Mountain Shorts, M	394913	663155	59.55%
Women's Mountain Shorts, L	286059	484179	59.08%
Sport-100 Helmet, Blue	1048446	1775081	59.06%
Half-Finger Gloves, S	191760	324906	59.02%
Sport-100 Helmet, Red	1191402	2019021	59.01%
Classic Vest, S	170581	289213	58.98%
HL Road Tire	268136	455456	58.87%
Hitch Rack - 4-Bike	305326	519211	58.81%
All-Purpose Bike Stand	201064	342664	58.68%
Half-Finger Gloves, M	152834	260780	58.61%
Sport-100 Helmet, Black	1141330	1948695	58.57%
Classic Vest, L	157319	268943	58.50%
Half-Finger Gloves, L	166721	285098	58.48%
Bike Wash - Dissolver	116008	198745	58.37%

Create Pivot Tables (Task 11)

Stacked Column Chart: Revenue by Product Category per Year

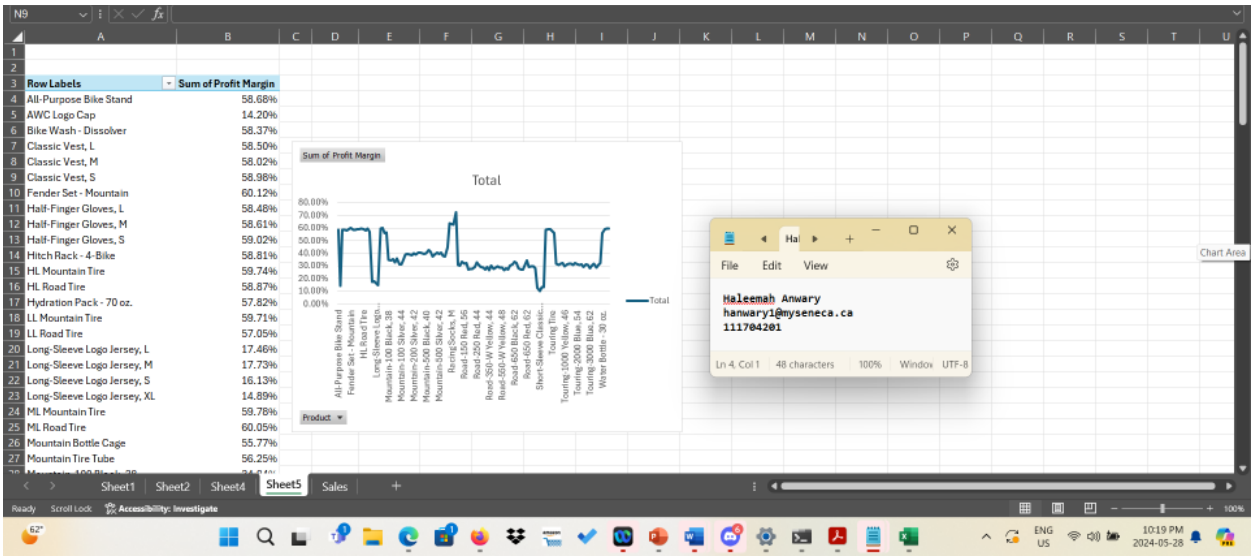
Why This Chart: Stacked column charts are excellent for showing part-to-whole relationships, allowing viewers to easily see the total revenue per year as well as the contribution from each product category within those totals.



Sum of Revenue	Column Labels			
Row Labels	Accessories	Bikes	Clothing	Grand Total
2011		8964888		8964888
2012		9175983		9175983
2013	3384215	9858787	1997035	15240037
2014	4293592	7611243	2247889	14152724
2015	3285954	14799083	1938954	20023991
2016	4154231	11372150	2187004	17713385
Grand Total	15117992	61782134	8370882	85271008

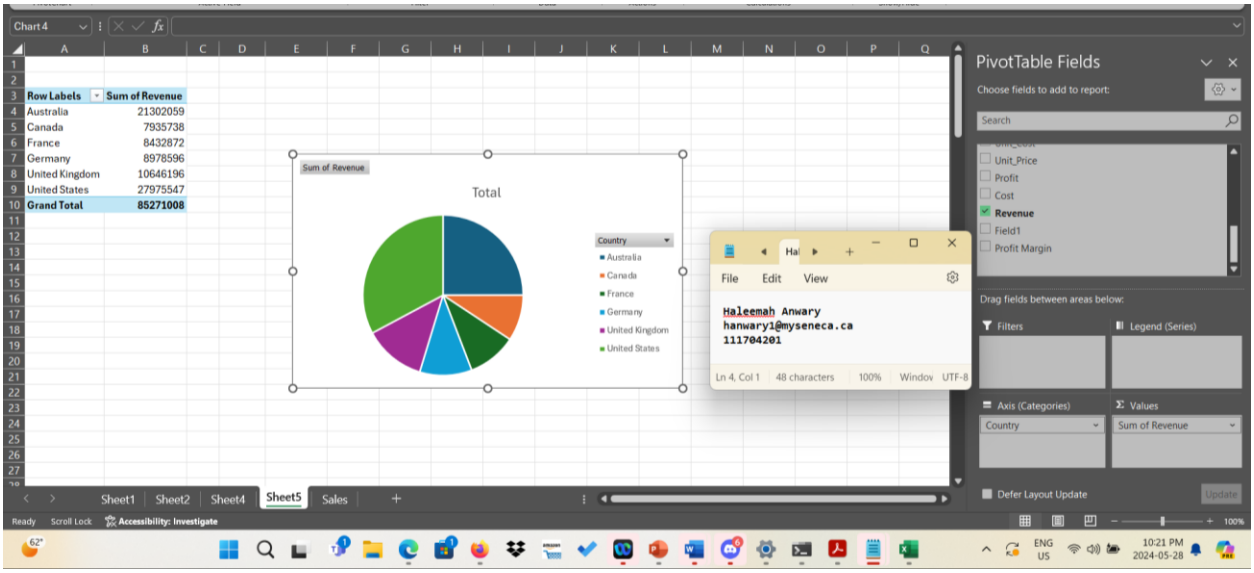
Line Chart: Profit Margin by Product

Why This Chart: Line charts are ideal for displaying data that changes over time or, in this case, across categories. It clearly shows trends in profit margin across different products, highlighting fluctuations or patterns in profitability.



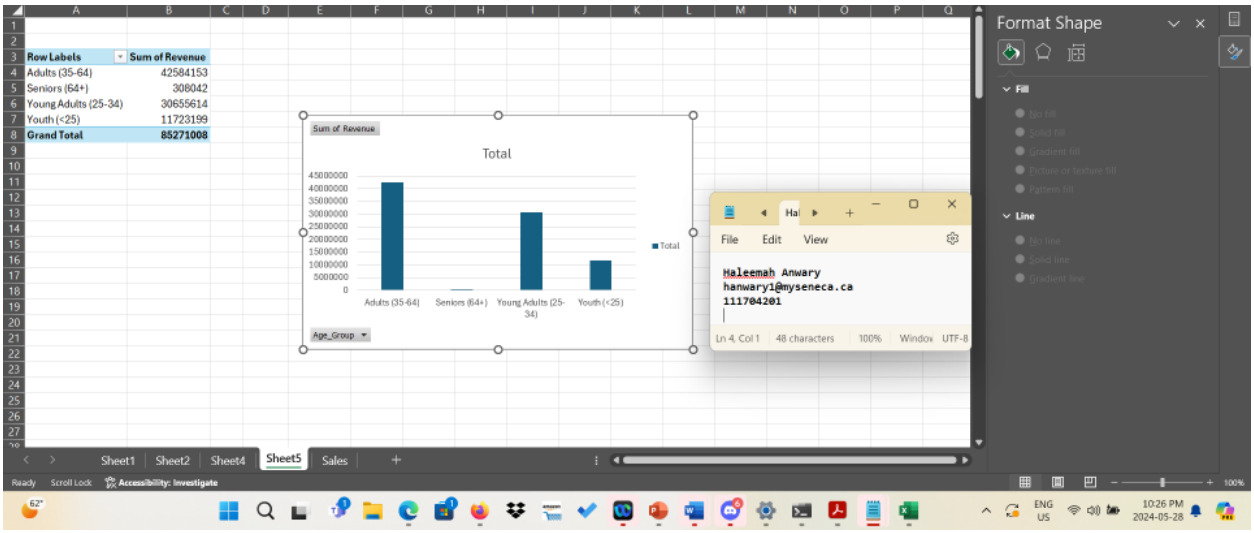
Pie Chart: Sales by Country

Why This Chart: Pie charts are useful for showing the proportion of parts to a whole, making it easy to compare the revenue contribution of each country to total sales. This helps quickly identify which countries are the largest markets.



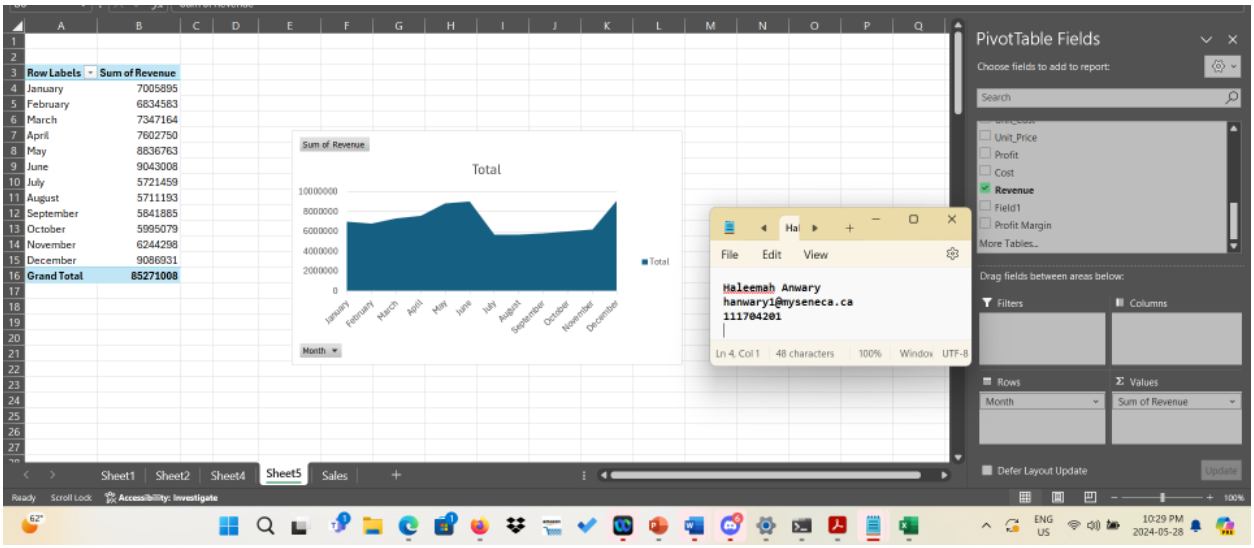
Bar Chart: Highest Sales by Age Group

Why This Chart: Bar charts are straightforward and effective for comparing quantities across different categories. In this case, it allows for a clear comparison of sales across different age groups, highlighting which age group generates the most revenue.



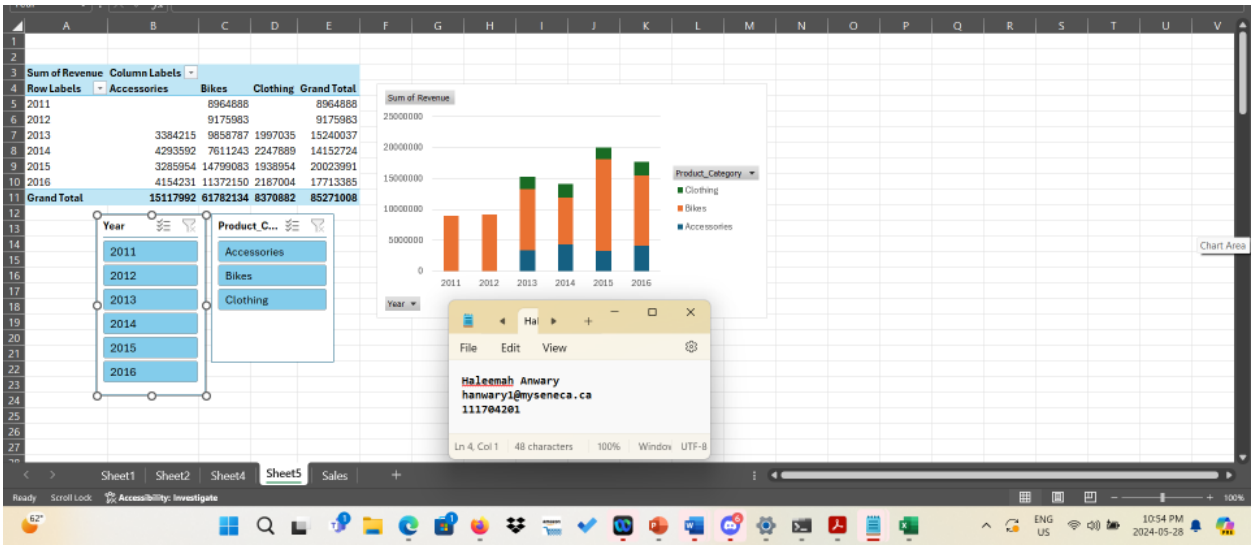
Area Chart: Monthly Revenue Trends

Why This Chart: Area charts are great for illustrating the cumulative magnitude of values over time (or sequence), showing the trends in revenue across months. This can highlight peak sales periods and help in forecasting.



Using Slicers to Filter Data (Task 12)

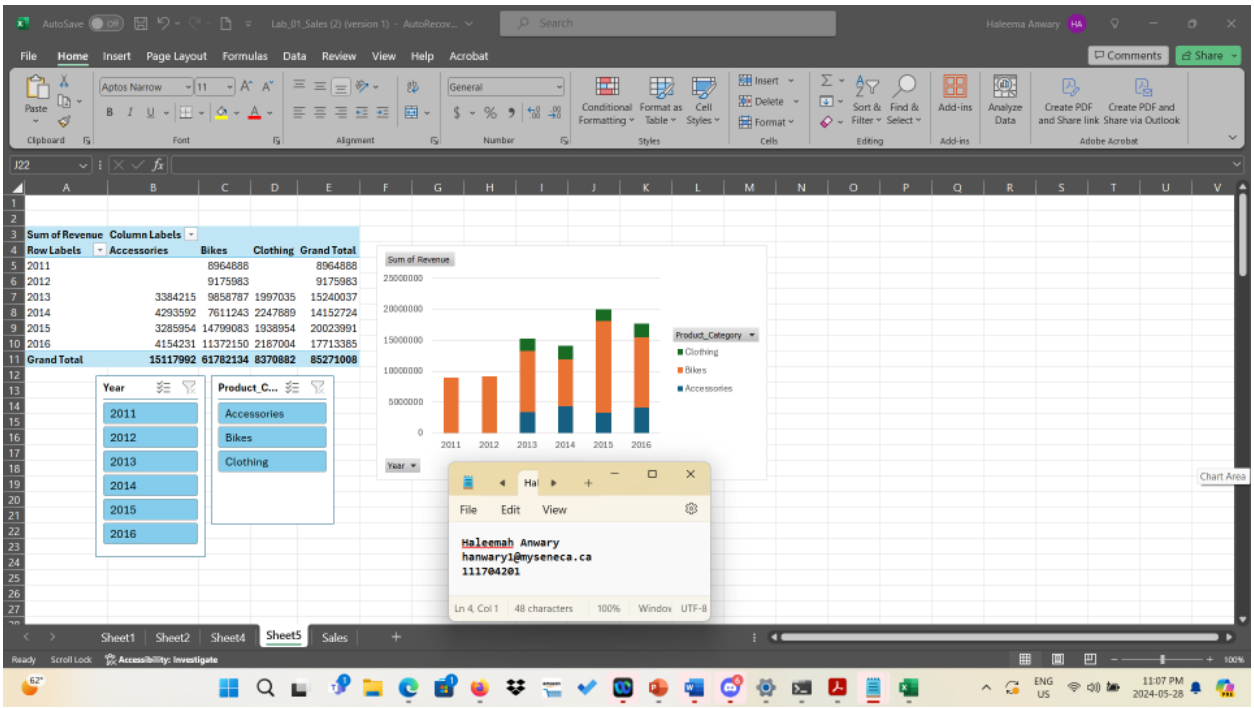
I used slicers for chart 1 which was the stacked bar graph. I added slicers for Year and Product_Category.



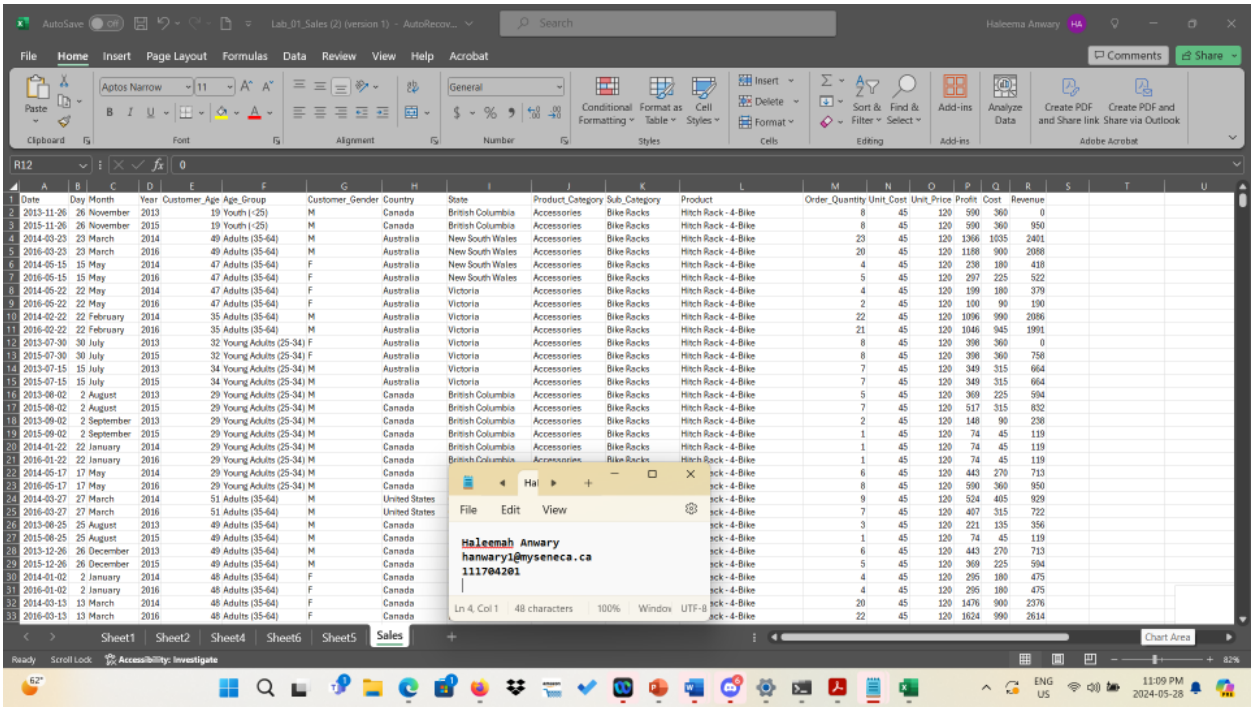
Controlling Charts with Report Connection (Task 13)

Updating Data and Refreshing the Dashboards (Task 14)

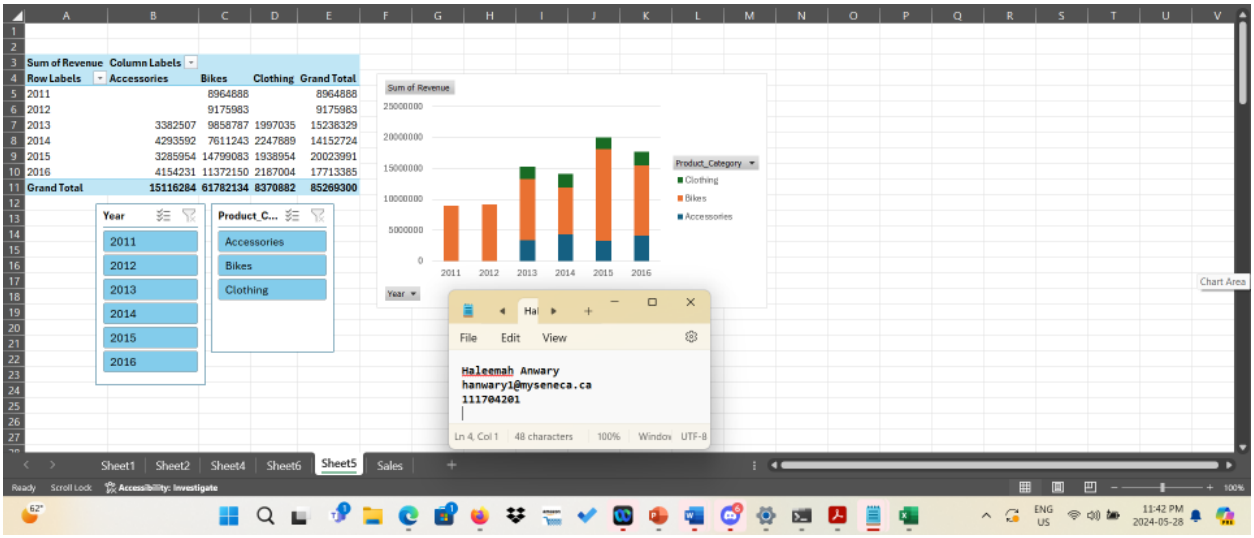
Here I have a pivot table which shows that the revenue for 2013 for accessories is $3384215.



I changed some values for accessories for 2013 to $0 on the base sheet of the pivot table. Went to the 'Data' at the top and clicked 'Refresh All.'



Now our pivot table also updated because it's linked. The new total revenue for accessories in the year 2013 is $3382507.



The impact of Refreshing the Dashboard

Using the "Refresh All" feature in Excel ensures that any updates made to the source data are immediately reflected across all related PivotTables and charts within the dashboard. This functionality guarantees that visual elements display the most current data, maintains consistency and accuracy across reports, and allows users to see changes in real-time. Refreshing the dashboard is crucial for preventing discrepancies that could lead to erroneous decisions and for ensuring that visualizations like charts adapt dynamically to new data.

Learning Outcome Report (Task 15)

Introduction

Participating in Labs 1 and 2 has provided me with a comprehensive understanding of data analytics using Microsoft Excel. These exercises enhanced my abilities in handling, analyzing, and visualizing data, essential skills in today's data-driven environment.

Key Learnings

Lab 1 was centered around fundamental data analytics operations within Excel, focusing on the use of pivot tables, advanced formulas, and data manipulation tools. It equipped me with the ability to structure unorganized data into a coherent form for analysis and apply various analytical techniques to extract meaningful insights, including creating pivot tables to analyze complex datasets effectively and utilizing formulas such as IF, SUMIF, and XLOOKUP to sift through data based on specific criteria.

Building on the foundational skills from Lab 1, Lab 2 introduced more complex tasks involving data cleaning, visualization, and interactive dashboard creation. I learned techniques for ensuring data accuracy and consistency, crucial for reliable analysis. The lab emphasized the importance of visual presentation in data analytics, teaching me to not only analyze data but also to present it in an intuitive manner that can be easily understood by stakeholders. Creating and customizing charts to effectively visualize data and building interactive dashboards using PivotTables, PivotCharts, and slicers facilitated dynamic data exploration.

Transferability to Cybersecurity and Data-Driven Analytics

The skills developed through these labs are highly applicable to cybersecurity and data-driven analytics. Cybersecurity, much like any field within IT, relies heavily on data for threat detection, risk assessment, and decision-making. Data cleaning and preparation ensure that the data used for security analysis is accurate and free from errors, which is paramount in cybersecurity. The ability to analyze and interpret data using tools like pivot tables can directly translate to analyzing security logs and incident reports, helping to identify patterns that signify potential threats or breaches. Visualization skills are crucial for monitoring network activities and detecting unusual patterns quickly, enabling security professionals to communicate complex

data and findings in a comprehensible manner. Additionally, the dynamic nature of interactive dashboards allows for real-time data exploration, crucial in a rapidly evolving threat environment, enabling quick response and mitigation strategies.

Conclusion

The exercises and concepts mastered in these labs have laid a robust foundation in both analytical and visualization techniques using Excel. These skills not only enhance my capability to perform complex data analyses but also equip me to apply these techniques in cybersecurity, where data-driven strategies are increasingly becoming central to operations and strategic decision-making. This integration of data analytics into cybersecurity can significantly enhance an organization's security posture by enabling a more proactive and informed approach to threats and vulnerabilities.